MECH5200 Numerical Methods for PDEs : *Video 18: Finite Differences For Hyperbolic Equations*

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References and Acknowledgements

The following materials were used in the preparation of this lecture:

1. Tannehill, Anderson and Pletcher, *Computational fluid Mechanics and Heat Transfer*.
2. 16.920, lecture 8,9,10 Notes
3. Sethian, J.A., Level Set Methods – Evolving interfaces in Geometry, Fluid Mechanics, ...

The author of these slides wishes to thank these sources for making the current lecture.

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Linear Hyperbolic Problem – The Wave Equation

- The simplest *hyperbolic equation* is the linear wave equation:

$$\frac{\partial u}{\partial t} + C\frac{\partial u}{\partial x} = 0 \tag{1}$$

- With initial condition: $u(x,0) = u_0(x)$
- Boundary Conditions:
  - $u(0,t) = g_0(t) \qquad (\text{if } C > 0)$
    OR:
  - $u(0,t) = g_1(t) \qquad (\text{if } C < 0)$

Solution to the Wave Equation

- Let's examine time derivative of the solution u in a moving vs. stationary reference frame (Eulerian vs. Lagrangian) :

$$\left(\frac{D(u(x(t),t))}{Dt}\right)_{Lagrangian} = \left(\frac{\partial u}{\partial t}\frac{dt}{dt}\right)_{Eulerian} + \left(\frac{\partial u}{\partial x}\frac{dx}{dt}\right)$$

- But this becomes:

$$\left(\frac{D(u(x(t),t))}{Dt}\right) = \left(\frac{\partial u}{\partial t}\right) + C\left(\frac{\partial u}{\partial x}\right) = 0$$

- Our wave equation...

Solution to the Wave Equation

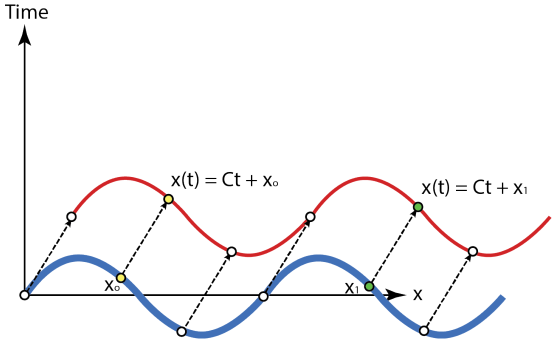
- This means, $\frac{Du}{Dt} = 0$. In other words, the solution u does not change if we follow/move with that point.
- Also $\frac{dx}{dt} = C$ & because of this, we also know that:

$$x(t) = Ct + \xi$$

- Is the line along which the observer is traveling. This is a line along which the PDE becomes an ODE.
- This line is called the characteristic line.

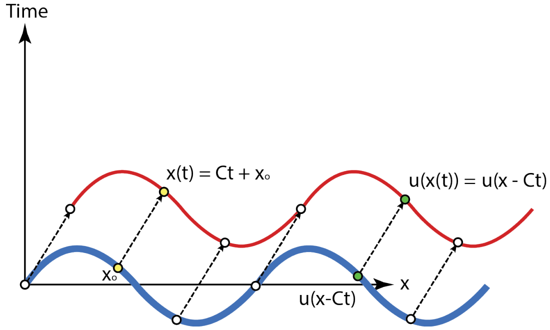
Let's Graphically Look at the Wave Equation

- If we assume an observer is moving on a curve $x(t)$ and is observing the solution u.



Solution to the Wave Equation

- The solution to the wave equation is: $u = \textit{constant}$ along the characteristic line.
- This means: $u(x, t) = u(x - Ct)$



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1. Linear Hyperbolic Problems – The Wave Equation

2. Model Problem and Finite Difference Solutions

Model Problem

- Let's try a model problem – and then use finite differences to solve it:
- Consider a unit domain governed by:

$$\frac{\partial u}{\partial t} + C\frac{\partial u}{\partial x} = 0$$

- With boundary conditions:

$$u(0,t) = u(1,t)$$

- and initial condition:

$$u(x,0) = sin(2\pi x)$$

- Notice that we have changed the problem slightly, so that the boundary conditions are now **periodic**.

Finite Differences – Domain Discretization

- How do we discretize this?
  - Discretize spatial domain with J nodes ($J-1$ equal intervals with size $\Delta x = \frac{1}{J-1}$)
  - Discretize time into N equal temporal intervals with size $\Delta t = \frac{T_{end}}{N}$
  - Write your finite difference expressions for this....
  - QUESTION: What are the most logical spatial and temporal approximations for the wave equation?

F-D approximation – Forward in time, backward in space

- Using forward derivative in time and backward derivative in space give us:

$$u_t + Cu_x = 0 \simeq \frac{\hat{u}_j^{n+1} - \hat{u}_j^n}{\Delta t} + C\frac{\hat{u}_j^n - \hat{u}_{j-1}^n}{\Delta x} = 0$$

- This can be re-arranged as follows:

$$\hat{u}_j^{n+1} = \hat{u}_j^n - \frac{\Delta t C}{\Delta x}(\hat{u}_j^n - \hat{u}_{j-1}^n) = 0$$

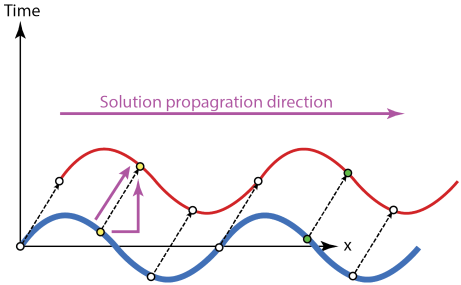
- This yields the Courant Number:

$$Courant\# = \frac{C\Delta t}{\Delta x} = \frac{D_{propagated}}{Grid \;\; spacing}$$

- The **Courant Number** is critical to the solution of hyperbolic equations.

Let's code this up in MATLAB and test it

- The **Courant Number** is critical to the solution of hyperbolic equations.
- It is the ratio of how far the solution propagates as a percentage of the grid spacing.



Assessing the MATLAB Example – CFL condition

- The *CFL* condition:
  - The mathematical zone of dependence must be contained in the numerical zone of dependence.
  - This usually means that the **Courant Number** must be less than a value of unity.

What have we learned?

- Hyperbolic equations require spatial derivatives to "obey" the propagation of information.
- Forward in time, backward in space (used in this example)
- Courant number describes how far the numerical solution propagates as a function of the grid size
  - Must be between zero and one.
  - This means that information propagates reasonable distances.
- This is the end of Finite Difference Section